

Why Are Air Conditioners Still Hard to Come By in India?

Zhihu

One-sentence answer: because India's level of industrial development is roughly where China was 20 years ago. As India's air-conditioner and compressor capacity expands by leaps and bounds, India's AC ownership will rise rapidly within a few years.

Through joint ventures with Japanese firms and the like (including compressor manufacturing technology obtained from Japan and the United States — Midea's technology came from Japan, Gree's from Japan and America; by the current Chinese attitude toward India, the behaviour of the Japanese and American firms would count as "aiding the enemy"), China gradually built up its domestic AC production capacity in the 1990s. From the late 1990s into the early 2000s, China experienced a great surge in AC capacity, which led to oversupply and finally to a drastic fall in AC prices:

In the 2002 cooling year, AC output reached 28.61 million units, but domestic market demand was only 15.9 million; although exports surged to 8 million, nearly 5 million units of surplus capacity remained. Small factories took the lead in launching price wars and vicious competition, at one point producing a chaotic scramble to trade price for volume. The most emblematic event was AUX's release in 2002 of its "White Paper on Air Conditioner Manufacturing Costs," breaking down and publicising the cost structure of ACs. Hisense, Kelon, Changhong, and others also cut prices one after another. From 2001 to 2004, the average price of domestically sold ACs fell from 2,835 yuan to 1,600 yuan, an annual decline of 17.4%. Source: <https://mp.weixin.qq.com/s/8NbFpPaG2XUjFZO6Zc9pmA>

In 2005, China's per-capita annual disposable income was 6,385 yuan; urban per-capita disposable income was around 10,000 yuan; migrant workers earned on average 861 yuan a month (only a little over 600 in 2001). Even after the price cuts, most city dwellers in China still had to spend nearly two months of wages to buy an air conditioner. Rural people's annual disposable income was only a little over 3,000 yuan — they basically could not afford ACs at all.

In 2025, India's AC demand is about 15 million units, comparable to China's in the early 2000s, but domestic output is only about 6 million. Because of this year's abnormal weather, India's AC market may reach about 20 million units. Contrary to these pre-capitalist views on Zhihu, a supply shortage is not a problem for capital at all — it is a vast business opportunity. Shortage means high prices; high prices mean high profits. With AC technology already mature today, this constitutes no bottleneck to capital accumulation — on the contrary, it stimulates accumulation. The notion that "the tropics

cannot industrialise" is pure fantasy. How much fear of India's rise is carried in such Chinese talk is worth pondering.

In fact, once an economy reaches a certain stage of development, home appliances become a hard necessity — air conditioners and refrigerators especially. Just as China's huge demand in the 1990s drew massive capital into the AC market and rapidly expanded capacity, India is now going through the same process. To exchange its market for production capacity, the Indian government has deliberately raised tariffs on home appliances, keeping AC prices in India generally high.

At present, an ordinary T1 air conditioner in India (rated for up to 43°C) sells on average for a little over 2,300 yuan; a tropical T3 unit (rated for up to 52°C) averages nearly 3,000 yuan. With installation, the total comes to about 3,000–4,000 yuan. With India's rapid industrialisation in recent years, workers' wages have also been rising, and southern India has even experienced a "migrant labour shortage." Indian migrant workers' wages currently stand at over 1,000 yuan a month for unskilled work such as porters and general site labour, and above 1,500 yuan for skilled trades such as bricklaying. In the southern Indian states wages are higher still; in Kerala, the highest, monthly wages commonly exceed 1,500 yuan. So although ACs are expensive, on balance an Indian worker can still buy one with two to three months' wages — much like China in the early 2000s before the price collapse, or even somewhat better; a little worse than after the big price cuts, to be sure.

India's AC capacity is expanding rapidly; the industry commonly expects self-sufficiency by 2028, although the core compressor components will still have to be imported. At present, India has not broken through in indigenous compressor technology, nor can it simply buy technology everywhere as China could in the 1990s, so the expansion of domestic capacity depends mainly on Japanese and Korean firms. A number of Japanese and Korean companies are currently planning to expand production in India, making ACs for the Indian environment:

Daikin will build a new AC production plant in India, targeting around 2027. Daikin is currently producing home air conditioners at two sites in India, with annual capacity of about 3 million units. Daikin will select land in India and raise capacity to 5–7 million units, more than 70% above the current level. Kanwal Jeet Jawa, the director overseeing Daikin's India and Africa operations, predicts: "AC penetration will rise to 12% in the next few years. India's AC market will reach 24 million units by 2029, a big increase from 7 million in 2019." Source: https://mp.weixin.qq.com/s/aDBMnC_roApzJBSP5MguoQ

On 6 February, Indian media reported that Mitsubishi Electric's brand-new AC and compressor plant in Chennai, Tamil Nadu, had officially gone into production. The plant, an investment of about 23 million dollars and covering about 210,000 square metres, is planned for an annual capacity of

300,000 home AC units and 650,000 compressors, and will create more than 2,100 direct and indirect jobs. Source: <https://mp.weixin.qq.com/s/53tbd6qOL3JCZPjzzti1MQLG>

LG Electronics India has announced the construction of its third manufacturing plant in the state of Andhra Pradesh, planning to invest about 600 million dollars over the next four years to attract supporting industries to the region. The new plant reportedly covers 1 million square metres, with a total floor area of 220,000 square metres, and will begin producing air conditioners from the end of next year; from 2029 it will sequentially start production lines for washing machines, refrigerators, and AC compressors. Annual capacities: 800,000 refrigerators, 850,000 washing machines, 1.5 million ACs, and 2 million AC compressors. Source: <https://mp.weixin.qq.com/s/aC7BSMMnHnjTfrLUoPKHTQ>

In addition, the Japanese firm Nidec (with technology acquired from a Brazilian company) is investing in a large compressor plant in India for refrigerators:

Nidec plans to invest 120 million dollars in a new 55,000-square-metre plant in India to produce Embraco-brand commercial and residential refrigeration compressors and electronic components. The plant is located in Chhatrapati Sambhajnagar (formerly Aurangabad), Maharashtra, with an annual capacity of 6 million units, and is scheduled to begin production in the second half of 2026. Once fully operational, it is expected to create 750–1,000 direct jobs, strengthening the local labour pool and its technical expertise in refrigeration solutions. Source: <https://mp.weixin.qq.com/s/DEwds8LDMhGRxlQJLkmOYQ>

Indian domestic AC plants — Tata's Voltas, for example — are still unable to make compressors and must import them from China; their expansion plans still depend on China. However, Voltas intends to build a compressor plant in Chennai with an annual capacity of 2 million units, planning to buy Japanese/Korean or Chinese technology to localise compressor production. As things stand, buying Chinese technology may prove difficult — it will probably have to buy Japanese.

On the whole, India's AC supply has greatly stimulated investment by domestic and foreign enterprises. Even if Chinese firms do not actively form joint ventures with India (the so-called pig-butcher), India, leaning on its rapidly expanding market, will most likely still be able to buy Japanese and Korean technology through joint ventures, and then, on the basis of digesting and absorbing the mature imported technology, achieve independent innovation. This is basically the same course Chinese enterprises followed from the 1990s to the 2000s, though a bit rockier. Once the great expansion of India's domestic capacity is complete, AC prices in India are expected to fall further, while wages rise somewhat more; the combination of the two will bring the air conditioner to every corner of the country.

In fact, the places in India where temperatures exceed 43°C are concentrated in the north, while the five most economically developed southern states and two western states are not so hot — being

coastal, they are humid, but an ordinary AC cools them fine. Also counter-intuitively, India's overall warming is lower than the global average; Chen Jing has an article on Guancha.cn discussing this. Chen Jing, as everyone knows, is no India cheerleader. The article is quite interesting; those interested may look it up.

On 11 April 2025, Science published an article titled "A global warming 'hole' where you'd least expect it." Since 1901, India's average annual temperature has risen by less than 0.7°C, only about half the global rate, forming a notable "warming hole." Source:
<https://mp.weixin.qq.com/s/b3b15qLAlPPtAskujWt56w>

As for the electricity problem that Chinese people constantly harp on as a check on India's development — India is also rapidly filling this short board. People may doubt the statistical calibre of the Indian government's figures, but India's power generation (especially solar), transmission, and appliance sales are all increasing, which reflects real and sustained progress in India's electricity sector. India plans to reach 2,000 kilowatt-hours of electricity per capita by 2030 — the level China reached in 2006.

1. Meeting record demand: India successfully met a maximum power demand of 242.49 GW in FY 2025-26. 2. Power shortages greatly reduced: thanks to the large increase in generation and transmission capacity, the national energy shortage rate fell to 0.03% in FY 2025-26, a marked improvement over 4.2% in FY 2013-14. 3. Per-capita consumption up: India's per-capita electricity consumption surged to 1,460 kWh in 2024-25, a 52.6% increase (503 kWh) over 957 kWh in 2013-14. 4. Supply improved: average power supply hours in rural areas have risen from 12.5 hours in 2014 to 22.6 hours, while urban areas now enjoy up to 23.4 hours (22.1 in 2014), reflecting a substantial improvement in the reliability and coverage of electricity services. Source:
<https://mp.weixin.qq.com/s/b1jB8dx1a0oF9257S42idg>

When observing India, the most important thing is to analyse it in terms of China's development level of 20 years ago. As for why India is said to be at China's 2005 level, see my detailed analysis: "Is India Really Underestimated Domestically?"

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